

E7 Introduction to Computer Programming for Scientists and Engineers

Discussion week 8

A brief introduction to linear algebra

The pillars of applied mathematics

- **Calculus** ... the study of change over space and time.
- **Linear algebra** ... the study of linear transformations.
- **Statistics** ... the study of uncertainty.
- **Optimization theory** ... the study of decision making.
- **Numerical methods** ... how to solve problems with a computer.

Basic concepts in linear algebra

- vector space
- vector
- matrix
- linear transformation

- A vector space is a collection of objects (called vectors) that

a) can be **added** together to produce another vector,

$$a, b \in V \quad \Rightarrow \quad a + b \in V$$

b) can be **multiplied by a scalar** to produce another vector,

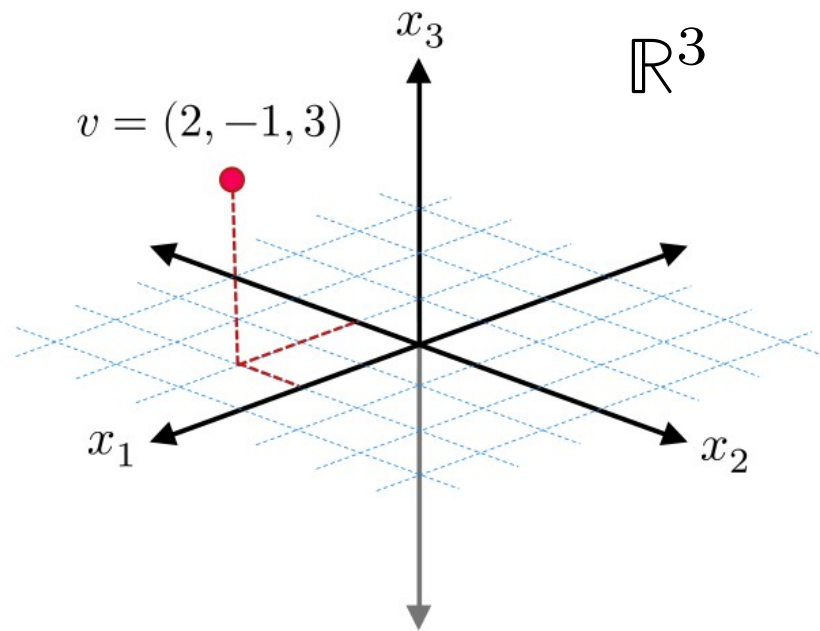
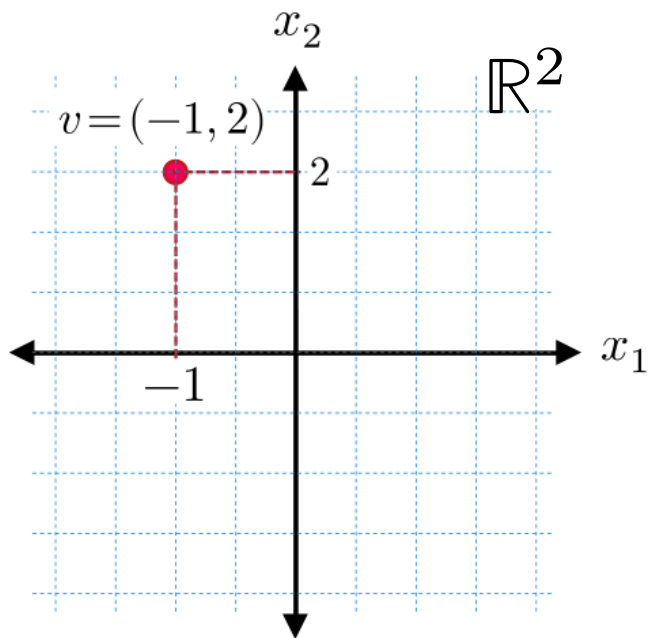
$$\begin{array}{l} a \in V \\ \alpha \in \mathbb{R} \end{array} \quad \Rightarrow \quad \alpha a \in V$$

Examples of vector spaces

- The set of all polynomials of a real variable of degree two or less.
- Numpy arrays of a given shape, with dtype=float.

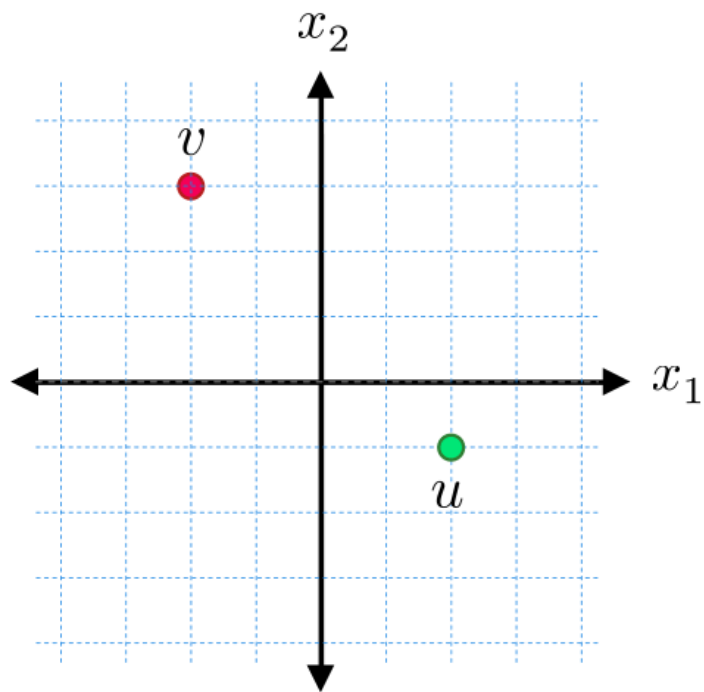
The most important vector spaces: $\mathbb{R}^n$

- For each integer $n \geq 0$, $\mathbb{R}^n$ is a vector space.
- Its vectors are of the form: $v = (v_1, v_2, \dots, v_n)$



Vector addition and scalar multiplication in $\mathbb{R}^2$

With $v = (-2, 3)$ and $u = (2, -1)$, find $u + v$ and $2u$.



Matrices

An m -by- n matrix M (denoted $M \in \mathbb{R}^{m \times n}$) is a grid of numbers with m rows and n columns.

$$M = \begin{bmatrix} m_{11} & m_{12} & m_{13} & m_{14} \\ m_{21} & m_{22} & m_{23} & m_{24} \\ m_{31} & m_{32} & m_{33} & m_{34} \end{bmatrix} \in \mathbb{R}^{3 \times 4}$$

Multiplying a matrix by a vector

- A matrix $M \in \mathbb{R}^{m \times n}$ can be “multiplied” by a vector $v \in \mathbb{R}^p$ when $n = p$.
- That is, the number of columns in M matches the number of items in v .

Example: Can M and v be multiplied?

- $M = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$, $v = (0, 4, 1)$

- $M = \begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$, $v = (3, 2, 1)$

How to multiply a matrix by a vector

- With $M \in \mathbb{R}^{m \times n}$ and $v \in \mathbb{R}^n$, their product Mv is a vector in $\mathbb{R}^m$.
- The i 'th element of Mv is the “dot product” of the i 'th row of M with v .
- A *dot product* of two vectors is the sum of their element-wise product.

Example

$$\begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \\ a_3 \end{bmatrix}$$

=

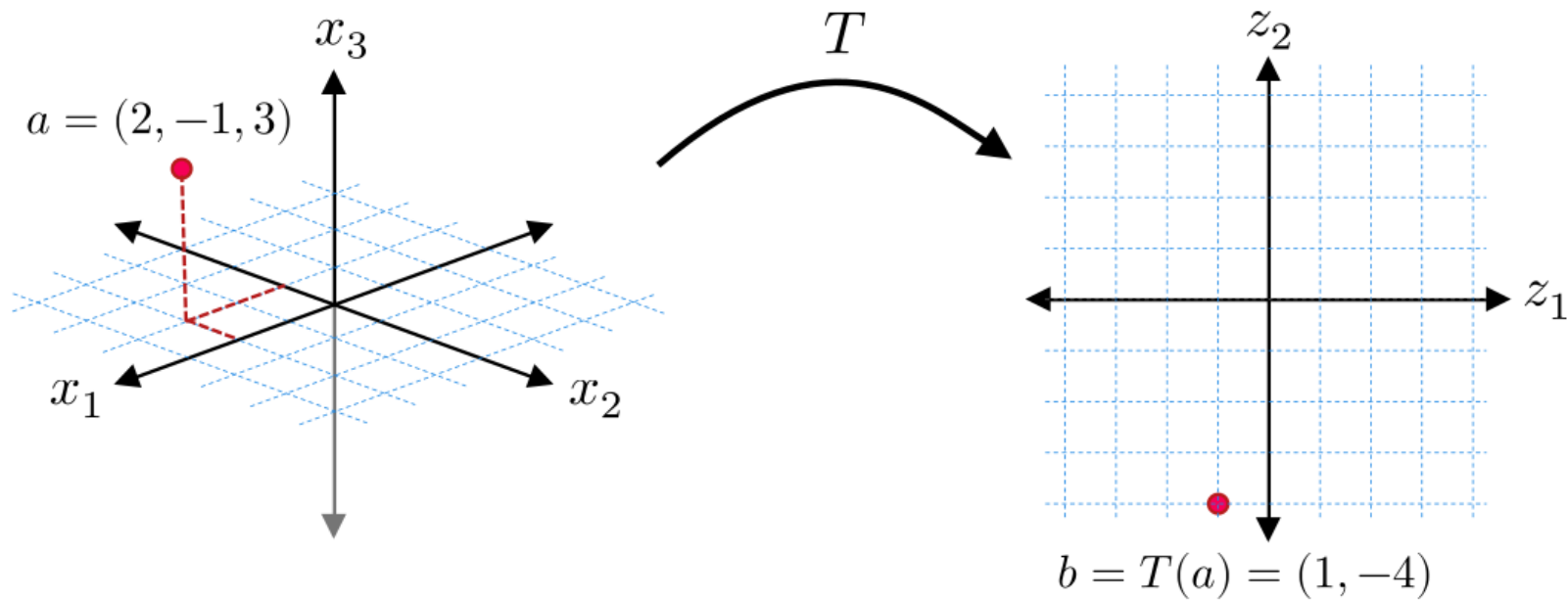
Linear transformations

A *linear transformation* T is function from a vector space V to a vector space W with the property

$$T(u + \alpha v) = T(u) + \alpha T(v) \quad \forall \alpha \in \mathbb{R}, \forall u, v \in V$$

Example $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$ maps each $a = (a_1, a_2, a_3)$ to a point $b = (b_1, b_2)$ with

$$T : \begin{cases} b_1 = 3a_1 + 2a_2 - a_3 \\ b_2 = a_2 - a_3 \end{cases}$$

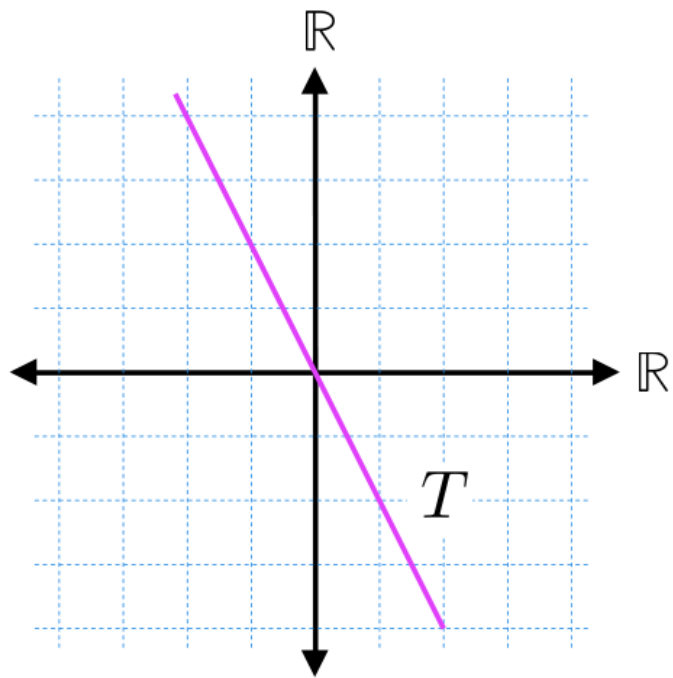


Show that this T is a linear transformation.

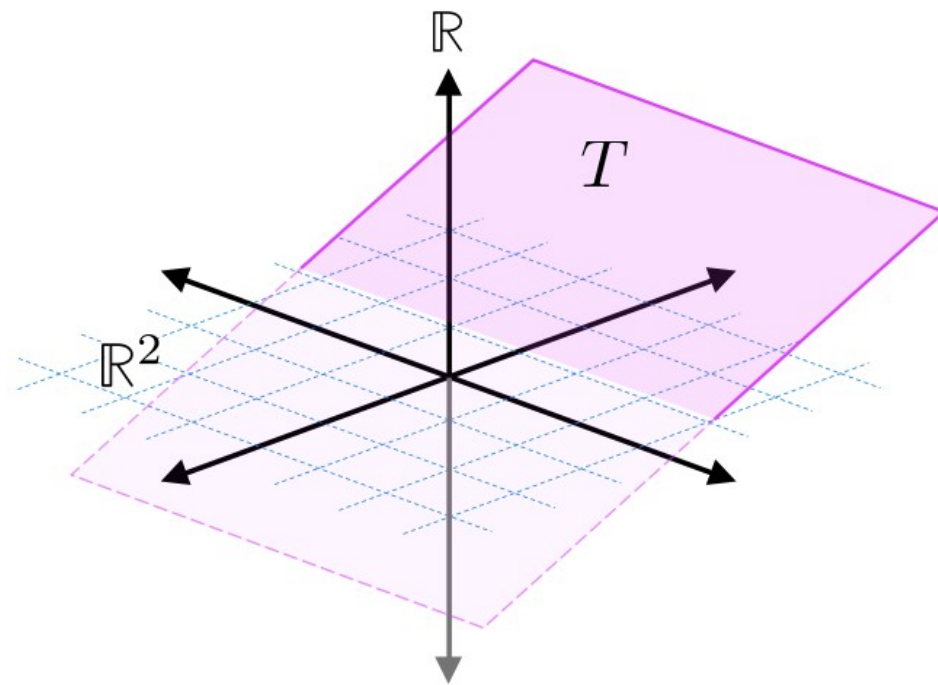
Proof:

Low dimensional linear transformations as a single plot

$$T : \mathbb{R} \rightarrow \mathbb{R}$$



$$T : \mathbb{R}^2 \rightarrow \mathbb{R}$$



Matrix representation theorem

For every linear transformation $T : \mathbb{R}^n \rightarrow \mathbb{R}^m$, there exists a matrix $M \in \mathbb{R}^{m \times n}$, such that

$$T(u) = Mu \quad \forall u \in \mathbb{R}^n$$

(*) Technical caveat: This is true only for “finite dimensional” vector spaces, but this is beyond our scope.

Example

Find the matrix M corresponding to the following linear transformation T .

$$T : \begin{cases} b_1 = 3a_1 + 2a_2 - a_3 \\ b_2 = a_2 - a_3 \end{cases}$$

Applications of linear algebra

- Linear transformations are very common in science and engineering.
 - Linear dynamical systems
 - Statistics and data science
 - Image processing and 3D graphics
 - Electrical circuit analysis
 - etc.
- The study of linear transformations (i.e. linear algebra) includes
 - matrix properties: rank and determinant
 - matrix decompositions: spectral (eigenvalues), singular values (SVD), etc.
 - matrix inversion: Moore-Penrose inverse